

Congestion-Aware Joint Charging-Station Placement, Capacity, and Swarm-UAV Trajectory Design for Age-Optimal Data Collection

Phuc Hao Do, Tran Duc Le, Truong Duy Dinh, and Van Dai Pham

Abstract—Unmanned aerial vehicle (UAV) swarms can keep Internet of Things data fresh, but energy-limited UAVs must recharge at a small number of shared charging stations, which then become a contended resource. We study the joint design of charging-station placement, port capacity, swarm trajectory, and UAV-to-station assignment to minimize the network peak Age of Information (AoI). A key observation is that the charging system is a closed population of only M UAVs, so it must be modeled as a finite-source (machine-repair) queue rather than an open $M/M/c$ queue. A discrete-event simulation confirms the finite-source model within about 4%, whereas the open queue overpredicts the waiting time by roughly $6\times$ and distorts the design decisions. We prove two structural results: a congestion threshold that makes the peak AoI non-monotone in the swarm size, with a provisioning law linking the optimal swarm size to the charging capacity, and a placement inversion showing that the coverage-optimal placement is AoI-suboptimal under contention. We then develop a block-coordinate solver with a GPU close-enough traveling-salesman trajectory optimizer. Across multi-seed experiments, the proposed method reduces the peak AoI by up to 12.2% over a strong single-station baseline, with each component isolated by an ablation and confirmed by a Wilcoxon signed-rank test ($p < 10^{-4}$). We also delineate the regime in which a single pooled station is already competitive.

Index Terms—Age of Information, unmanned aerial vehicles, UAV swarm, data collection, charging-station placement, finite-source queue, trajectory optimization, Internet of Things.

I. INTRODUCTION

Many Internet of Things applications depend on data that is not only correct but also fresh. In precision agriculture, environmental monitoring, disaster response, and industrial inspection, a control loop or an operator acts on the most recent sensor reading, so a stale reading can be worse than no reading at all. Ground sensors in these settings are often sparse, spread over a wide area, and far from any gateway,

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The source code and data that reproduce all results are available at <https://github.com/haodpsut/swarm-uav-aoi-congestion>.

which makes continuous fresh collection difficult with fixed infrastructure alone.

Unmanned aerial vehicles (UAVs) are attractive data collectors for such deployments: they fly close to ground sensors, establish strong line-of-sight links, and reach areas without fixed infrastructure. When the collected data is time-sensitive, the relevant metric is the Age of Information (AoI) [1], that is, the time elapsed since the freshest delivered update was generated. Keeping the network AoI low requires a UAV to revisit each sensor often, which conflicts with the limited battery of a rotary-wing UAV: the platform must periodically leave the field to recharge at a charging station.

A single UAV can only keep a small area fresh, so practical deployments use a *swarm*. A swarm sharing a small number of charging stations creates a problem that a single UAV never faces: the stations become a *contended* resource. When several UAVs need to recharge at the same time, they compete for a limited number of charging ports and must wait, and this waiting time inflates the AoI. The system designer must therefore decide jointly where to place the charging stations, how many ports to install at each, how the swarm should fly, and which UAV recharges where.

A. Limitations of prior work

Existing studies address parts of this problem but not the whole. The closest work jointly optimizes charging-station placement and trajectory, but for a *single* UAV, so there is no contention [2]. Multi-UAV studies that model charging either assume a single fixed facility [3], use mid-flight laser charging with fixed emitters and no queue [4], or use mobile ground vehicles that follow the UAVs and thereby remove the fixed-station contention [5], [6]. Swarm-AoI studies that ignore energy altogether [7] cannot capture recharging at all. A separate line models UAV charging stations as a queue with waiting time [8], but without any AoI objective, trajectory, or placement. A naive queueing model is misleading here: the charging system is a *closed* population of only M UAVs, so an open $M/M/c$ queue substantially overpredicts the wait, as we show against a discrete-event simulation.

B. Contributions

- We formulate the first joint charging-station *placement*, port *capacity*, swarm *trajectory*, and *assignment* problem for peak-AoI fresh data collection with a contended,

multi-server charging resource, as a mixed-integer non-linear program (Section III).

- We model the charging stations as a *finite-source* (machine-repair) queue and validate it against a discrete-event simulation. The finite-source model matches the simulation within about 4% and is a conservative upper bound on the real wait, whereas the open $M/M/c$ over-predicts by about $6\times$ and distorts the design decisions.
- We prove two structural results (Section IV): a *congestion threshold* that makes the AoI non-monotone in the swarm size, with a provisioning law linking the optimal swarm size to the charging capacity, and a *placement inversion* showing that the coverage-optimal placement is AoI-suboptimal under contention.
- We design a block-coordinate solver with a GPU close-enough-TSP trajectory optimizer (Section V) and evaluate it against single-UAV, single-station, coverage-optimal, and round-robin baselines (Section VI). The proposed method reduces peak AoI significantly, and we also identify the regime in which a single pooled station is already competitive. We further show that the greedy solver is within 0.1% of the exact optimum on small instances and that its advantage is robust to the collection radius, the charge time, and non-uniform sensor distributions.

The remainder of the paper is organized as follows. Section II reviews related work, Section III presents the model and problem, Section IV the structural analysis, Section V the solver, and Section VI the experiments. Section VII concludes.

II. RELATED WORK

A. Age of Information foundations

The Age of Information (AoI) was introduced in [1] as a destination-centric measure of data freshness, capturing the time elapsed since the generation of the most recently received update. The concept has since matured into a substantial body of theory, surveyed comprehensively in [9] and developed as a self-contained metric and analysis tool in [10]. Two threads of this literature are directly relevant to our problem. The first concerns sampling and transmission policy: [11] shows that, contrary to intuition, freshness is not always improved by updating as often as possible, and that an optimal policy may deliberately wait, while [12] analyzes how buffer and packet management shape the age process. The second concerns scheduling across contending sources: [13] derives age-optimal and near-optimal policies for broadcasting updates from many sources over a shared wireless channel. These foundations motivate our use of peak AoI as the design objective and, more subtly, they anticipate the tension our formulation confronts, namely that serving more sources with a shared resource (here, a shared charger) can raise age once contention is taken into account.

B. UAV communications, trajectory, and placement

Unmanned aerial vehicles have become a central platform for flexible wireless service, with the opportunities and open problems laid out in [14] and a broad tutorial treatment in [15]; the networking of cooperating aerial nodes is surveyed in [16].

Trajectory design is the dominant control lever in this setting: [17] jointly optimizes the paths and communication of multiple UAVs, and for data collection specifically [18] minimizes collection energy while [19] minimizes the flight time needed to gather data from a field of sensors. The quality of the air-to-ground link that these designs exploit is governed by the elevation-dependent line-of-sight probability characterized in [20] and by the aerial channel models standardized in [21], which we draw on when specifying the collection link. Two combinatorial structures recur in our own design. Deciding where to site a small number of charging stations is a facility-location problem, whose models and algorithms are treated in [22]. Planning a visiting tour in which a source is served whenever the UAV enters its communication range is a close-enough traveling salesman problem, formulated and solved exactly in [23] and heuristically in [24]. Our formulation couples these two structures, which are typically studied in isolation, to the freshness and charging dynamics described below.

C. AoI-oriented UAV data collection

Building on these foundations, AoI has become a standard objective for UAV-assisted data collection, where a UAV visits distributed sources and the design goal is to keep the collected information fresh. Single-UAV formulations establish the basic tradeoffs: [25] designs UAV-aided collection to maximize information freshness across a sensor field, and [26] minimizes average peak AoI in a UAV-assisted IoT network, jointly treating trajectory and update scheduling. In the swarm setting, [7] minimizes AoI across multiple cooperating UAVs using multi-agent reinforcement learning, demonstrating that coordinated trajectories can reduce peak AoI relative to uncoordinated flight; learning-based control of multiple UAVs is itself an active area, with resource allocation studied in [27] and deployment and movement design in [28]. That family of formulations, however, assumes flight endurance is unconstrained: it carries no energy state and no recharging model, so the depletion and replenishment dynamics that dominate long missions are absent. Closest to our objective is [2], which jointly optimizes charging station placement, UAV trajectory, and peak AoI, and which we adopt as our point of departure. That work is nonetheless single-UAV: with one vehicle there is no inter-UAV contention for a charger, no need for multi-port capacity, and no swarm assignment, precisely the coupling that our design targets.

D. Energy-limited UAVs and charging

A second line of work makes energy replenishment explicit. The battery lifetime of aerial nodes is a first-order operational constraint, as argued in [29], and wireless power transfer offers one replenishment mechanism whose trajectory and energy tradeoffs are optimized in [30]. Turning to shared charging infrastructure, [3] lets multiple UAVs share a limited charging facility and folds the resulting charge waiting time into the AoI objective, then optimizes it by deep reinforcement learning. The facility there is single and fixed in location, which effectively reduces contention to a single-server queue

and leaves station placement out of scope. Laser-based (LBD) charging is studied in [4], where charging sites are at fixed locations and are treated without queue contention, so neither where to place stations nor how many vehicles may wait is optimized. A different tactic uses mobile chargers: [5] and [6] employ ground vehicles that follow the UAVs to deliver energy on the move. Because the charger comes to the UAV, this removes the fixed-station contention that is central to our setting and does not address the placement of fixed, multi-port stations. Finally, [8] does model UAV charging stations as a contended queue and provides a waiting-time analysis, but it optimizes neither AoI nor trajectory nor station placement, so the queueing model is not coupled to any freshness or mobility decision. Beyond aerial platforms, a broad literature schedules the charging of vehicle and sensor fleets: auction-based time-slot allocation coordinates recharging across a multi-drone network in [31], decentralized protocols shift electric-vehicle charging in time to flatten aggregate demand in [32], and a mobile wireless charger is routed to replenish a sensor fleet in [33]. These dynamic-fleet-charging methods decide when and where each vehicle charges, yet they neither jointly place and size the shared stations nor couple the induced charging wait into an Age-of-Information objective for a UAV swarm, which is the focus of our formulation. When we make contention explicit, we model a station as a finite-source (closed) service system rather than an open queue. This places our analysis in the lineage of the machine-repair, or machine-interference, problem surveyed in [34] and treated within the classical queueing framework of [35], from which we adapt the finite-source multi-server results used in Section III.

E. Positioning and gap

Read together, these works each supply a subset of the ingredients our problem requires but never their combination. Swarm operation with AoI appears in [7] yet without charging; placement with AoI appears in [2] yet only for a single UAV; charge waiting is captured in [3] yet at a single fixed server without placement; and a contended charging queue is analyzed in [8] yet without AoI, trajectory, or placement. No prior formulation simultaneously carries all four of station placement, swarm operation, multi-server charging contention, and a peak-AoI objective. Our work closes this gap by jointly optimizing station placement, port capacity, swarm trajectory, and source-to-UAV assignment under a charging model that is a contended, multi-server, and finite-source (closed) queue, validated against discrete-event simulation. The finite-source aspect is essential rather than incidental: because the number of UAVs competing for a station is small and closed, modeling the queue as an open $M/M/c$ system overpredicts waiting time and thereby distorts both placement and capacity decisions, an effect we quantify and correct with the finite-source model detailed in Section III. This coupled formulation also yields two structural findings that a decoupled treatment cannot expose: a congestion threshold, beyond which peak AoI is non-monotone in swarm size, and a placement inversion, whereby coverage-optimal station placement becomes AoI-suboptimal once contention is accounted for.

TABLE I
SUMMARY OF NOTATION.

Symbol	Meaning
$K, \mathcal{K}$	number and set of ground sensors
M	swarm size (number of UAVs)
$\mathcal{S}, S_{\max}$	candidate sites, station budget
$x_s \in \{0, 1\}$	open a station at candidate site s
c_s, C_{tot}	ports at station s , total port budget
$\mathbf{q}_m(t)$	trajectory of UAV m
a_m	station assigned to UAV m
$A_k(t), \bar{A}$	AoI of sensor k , network peak AoI (objective)
V, H	cruise speed, altitude
$E_{\max}, \eta$	battery, reserve fraction
$\tau_{\text{fly}}, \tau_{\text{charge}}$	flight budget, charge time
T_{up}	operating time between charges (incl. travel)
N_s	UAVs assigned to station s
λ, μ	charge-request rate, service rate
W_q, t_{loop}	mean queue wait, sub-field revisit period

III. SYSTEM MODEL AND PROBLEM FORMULATION

The main symbols are summarized in Table I.

A. Network model

We consider a swarm data-collection system, illustrated in Fig. 1. A set $\mathcal{K} = \{1, \dots, K\}$ of ground sensors is deployed over a square field, with sensor k located at $\mathbf{w}_k \in \mathbb{R}^2$. A swarm of M rotary-wing UAVs, indexed by $\mathcal{M} = \{1, \dots, M\}$, flies at a fixed altitude H with horizontal position $\mathbf{q}_m(t) \in \mathbb{R}^2$ over discrete slots $t = 1, \dots, T$ of length δ_t . Each UAV holds a battery with state of charge $e_m(t) \in [0, E_{\max}]$. A set $\mathcal{S} = \{1, \dots, S\}$ of candidate charging-station sites is given, with site s at $\mathbf{b}_s \in \mathbb{R}^2$; the system may build a station and provision it with charging ports.

Freshness is measured by the Age of Information (AoI) [1]. Let $u_k(t)$ be the generation time of the freshest update from sensor k that has been delivered by time t . The instantaneous AoI is

$$\text{AoI}_k(t) = t - u_k(t), \quad (1)$$

which grows at unit rate and drops on each successful collection: when a UAV collects at time τ a packet generated at g , u_k jumps to g and $\text{AoI}_k(\tau^+) = \tau - g$. Following the base setting, the network objective is the worst-case peak AoI,

$$\bar{A} = \max_{k \in \mathcal{K}} \limsup_{t \rightarrow \infty} \text{AoI}_k(t), \quad (2)$$

which we seek to minimize. A UAV collects from sensor k while $\|\mathbf{q}_m(t) - \mathbf{w}_k\| \leq r_c$.

B. Rotary-wing energy

The charging cadence is not assumed constant: it emerges from how fast a UAV drains its battery in flight. For a rotary-wing UAV cruising at horizontal speed V , the propulsion power follows the closed-form model of Zeng *et al.* [36],

$$P(V) = P_0 \left(1 + \frac{3V^2}{U_{\text{tip}}^2} \right) + P_i \left(\sqrt{1 + \frac{V^4}{4v_0^4}} - \frac{V^2}{2v_0^2} \right)^{1/2} + \frac{1}{2} d_0 \rho s A V^3, \quad (3)$$

where P_0 and P_i are the blade-profile and induced power constants, U_{tip} the rotor tip speed, v_0 the mean rotor induced velocity in hover, and d_0, ρ, s, A the fuselage drag ratio, air density, rotor solidity, and disc area. Hovering during collection consumes $P(0) = P_0 + P_i$. The charge evolves as $e_m(t+1) = e_m(t) - P(V_m(t)) \delta_t$ while flying or collecting, and increases while docked at a charging port.

Only a usable fraction of the pack is spent between charges, keeping a reserve η , so $E_{\text{use}} = (1 - \eta)E_{\text{max}}$. Given a patrol loop of flight time $t_{\text{fly}}^{\text{loop}}$ and hover-collection time $t_{\text{col}}^{\text{loop}}$, the energy drawn per loop is $e_{\text{loop}} = P(V) t_{\text{fly}}^{\text{loop}} + P(0) t_{\text{col}}^{\text{loop}}$, so a UAV can fly $n_{\text{loop}} = \lfloor E_{\text{use}}/e_{\text{loop}} \rfloor$ loops before recharging, giving a physical flight budget between charges

$$\tau_{\text{fly}} = n_{\text{loop}} t_{\text{loop}}, \quad t_{\text{loop}} = t_{\text{fly}}^{\text{loop}} + t_{\text{col}}^{\text{loop}}. \quad (4)$$

A recharge-feasibility condition requires each UAV to retain enough charge to reach its assigned open station at all times: $e_m(t) \geq E_{\text{reach}}(\mathbf{q}_m(t), \mathbf{b}_{a_m})$.

C. Air-to-ground collection

Collection dwell is derived from a physical link rather than assumed instantaneous. With a line-of-sight, free-space channel of reference gain β_0 at 1 m, a sensor at horizontal distance d is seen at slant distance $\sqrt{H^2 + d^2}$, and the achievable rate is

$$R(d) = B \log_2 \left(1 + \frac{p_{\text{tx}} \beta_0}{(H^2 + d^2) N_0 B} \right), \quad (5)$$

with bandwidth B , sensor transmit power p_{tx} , and noise power spectral density N_0 . Collecting a status blob of L bits at the closest-approach distance therefore takes a dwell time $t_{\text{col}}(d) = L/R(d)$, which enters t_{loop} in (4) and hence the revisit period.

D. Finite-source charging queue

The load-bearing coupling of the swarm is contention for charging ports. A station s serving the N_s UAVs assigned to it with c_s parallel ports is *not* an open queue: only N_s UAVs exist, so the request rate falls as more of them are already at the station. The correct model is the finite-source (machine-repair) queue $M/M/c/N_s$ [35].

Each assigned UAV, while patrolling, requests a charge after a mean operating time, at rate $\lambda = 1/T_{\text{up}}$; each port serves at $\mu = 1/\tau_{\text{charge}}$. The operating time between two charge requests is the whole time a UAV spends away from the station in a cycle, so it must include the round-trip travel to the station and not the flight budget alone:

$$T_{\text{up}} = \tau_{\text{fly}} + \frac{2d(m, s)}{V}, \quad (6)$$

where $d(m, s) = \|\mathbf{q}_m - \mathbf{b}_s\|$. Omitting the travel term overstates λ and over-predicts the wait by roughly a factor of two.

Let $n \in \{0, \dots, N_s\}$ be the number of UAVs at the station. The birth-death chain has state-dependent rates

$$n \rightarrow n+1 : (N_s - n)\lambda, \quad n \rightarrow n-1 : \min(n, c_s)\mu, \quad (7)$$

where the up-rate $(N_s - n)\lambda$ vanishes as $n \rightarrow N_s$: this closed-population term is exactly why the queue cannot blow up. The stationary distribution is

$$p_n = p_0 \prod_{k=0}^{n-1} \frac{(N_s - k)\lambda}{\min(k+1, c_s)\mu}, \quad \sum_{n=0}^{N_s} p_n = 1. \quad (8)$$

The mean number waiting, the effective throughput, and, by Little's law, the mean queue wait are

$$L_q = \sum_{n=0}^{N_s} \max(0, n - c_s) p_n, \quad (9)$$

$$\lambda_{\text{eff}} = \sum_{n=0}^{N_s} (N_s - n) \lambda p_n,$$

$$W_q(N_s, c_s) = \frac{L_q}{\lambda_{\text{eff}}}. \quad (10)$$

Because the population is finite, no stability constraint (of the form $\rho_s < 1$) is needed: the wait *saturates* rather than diverging. This finite-source model replaces the open $M/M/c$ (Erlang-C) form, which treats arrivals as an inexhaustible stream and, for the cycling swarm, over-predicts the wait by roughly a factor of six. A discrete-event validation of the true cycling UAVs, reported in Section VI, confirms that (10) tracks the simulation to within a few percent and is a conservative upper bound on the real, near-deterministic wait.

E. Peak-AoI expression

For a UAV m that patrols a sub-field and leaves it to recharge, the worst-case revisit gap seen by its sensors is one patrol period plus one charging excursion:

$$\text{AoI}_m = t_{\text{loop}, m} + \frac{2d(m, s)}{V} + W_q(N_s, c_s) + \tau_{\text{charge}}. \quad (11)$$

The first term is the patrol revisit period, the second the round trip to the assigned station, the third the finite-source queue wait from (10) (the term that couples the swarm), and the last the charging time. The network peak AoI (2) is the maximum of (11) over the swarm. The coupling in W_q is the mechanism behind the non-monotone behavior of $\bar{A}$ in the swarm size M : more UAVs shrink the coverage term but lengthen the wait, up to the ceiling set by the finite population.

F. Joint optimization problem

We jointly optimize the placement decisions $x_s \in \{0, 1\}$ (build a station at s), the capacities $c_s \in \mathbb{Z}_{\geq 0}$ (ports per sta-

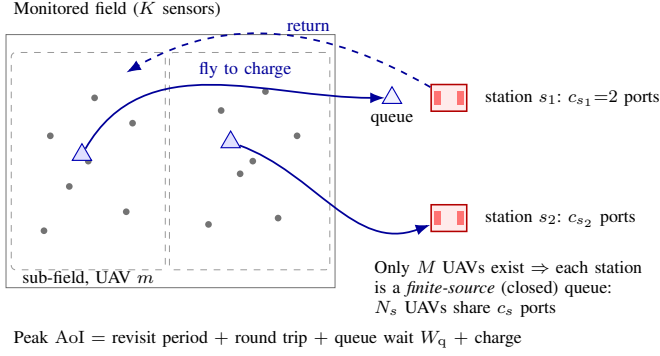


Fig. 1. System model: a UAV swarm patrols sub-fields to keep sensor data fresh, and recharges at a small number of shared, multi-port charging stations. Because only M UAVs exist, the charging stations form a *finite-source* (closed) queue.

tion), the UAV trajectories $\{\mathbf{q}_m(\cdot)\}$, and the station assignment $\{a_m\}$. The problem is

$$(P0) \quad \arg \min_{\mathbf{x}, \mathbf{c}, \{\mathbf{q}_m\}, \{a_m\}} \bar{A} = \max_{k \in \mathcal{K}} \limsup_{t \rightarrow \infty} \text{AoI}_k(t) \quad (12a)$$

$$\text{s.t.} \quad e_m(t) \in [0, E_{\max}], \quad \forall m, t, \quad (12b)$$

$$e_m(t) \geq E_{\text{reach}}(\mathbf{q}_m(t), \mathbf{b}_{a_m}), \quad \forall m, t, \quad (12c)$$

$$\text{AoI}_k(t) \text{ evolves as in (1), } \forall k, t, \quad (12d)$$

$$\sum_s x_s \leq S_{\max}, \quad \sum_s c_s \leq C_{\text{tot}}, \quad (12e)$$

$$c_s \leq C_{\text{port}} x_s, \quad x_s \in \{0, 1\}, \quad c_s \in \mathbb{Z}_{\geq 0}, \quad (12f)$$

$$\|\mathbf{q}_m(t+1) - \mathbf{q}_m(t)\| \leq V_{\max} \delta_t, \quad \forall m, t, \quad (12g)$$

$$a_m \in \{s : x_s = 1\}, \quad \forall m, \quad (12h)$$

$$\|\mathbf{q}_m(t) - \mathbf{q}_{m'}(t)\| \geq d_{\min}, \quad \forall m \neq m', t. \quad (12i)$$

Constraint (12c) is recharge feasibility, (12d) ties peak AoI to collection events through (11) and the queue wait (10), (12e) and (12f) enforce the station and port budgets, (12g) is UAV kinematics, (12h) restricts assignment to open stations, and (12i) is inter-UAV separation.

Problem (P0) mixes binary placement $\mathbf{x}$, integer capacity $\mathbf{c}$, continuous trajectories, and a combinatorial assignment, and its objective embeds the nonconvex finite-source wait (10) through the coupling (11). It is therefore a mixed-integer nonlinear program (MINLP) and NP-hard, which motivates the block-coordinate decomposition (trajectory, queue evaluation, capacity allocation, assignment, and outer placement) detailed in Section V.

IV. STRUCTURAL ANALYSIS

We now establish the two structural properties that separate the swarm regime from the single-UAV base case: the peak AoI is non-monotone in the swarm size (Proposition 1), and the AoI-optimal station placement inverts from coverage-driven to traffic-driven once the swarm contends for ports (Proposition 2). Throughout, we work with the per-UAV

peak AoI of the homogeneous single-station cell derived in Section III,

$$\text{AoI}(M) = \underbrace{t_{\text{loop}}(M)}_{c_1(M)} + \underbrace{W_q(M, c)}_{\text{contention}} + \underbrace{\frac{2 \text{dist}}{V} + \tau_{\text{charge}}}_{c_0}, \quad (13)$$

where M UAVs share one field and one station of c ports, $t_{\text{loop}}(M)$ is the patrol (revisit) time per UAV, $W_q(M, c)$ is the mean charging wait of the finite-source queue $M/M/c/M$ of Section III (population M , per-source operating rate $\lambda = 1/T_{\text{up}}$, port rate $\mu = 1/\tau_{\text{charge}}$), and c_0 collects the recharge round trip and the charging time, which do not depend on M .

A. Congestion threshold: AoI is non-monotone in swarm size

We first isolate the two competing forces in (13): the coverage term $c_1(M)$ that a larger swarm improves, and the wait $W_q(M, c)$ that a larger swarm degrades.

Assumption 1 (Coverage scaling): The patrol term $c_1(M) = t_{\text{loop}}(M)$ is positive, strictly decreasing, and convex in M , with $c_1(M) = \Theta(1/M)$.

The scaling is the standard sub-tour argument: M UAVs partition K sensors of a field of area $\mathcal{A}$ into M sub-fields of area $\mathcal{A}/M$ and K/M sensors each. The length of a near-optimal tour through n points uniformly spread over an area $\mathcal{A}'$ scales as $\Theta(\sqrt{\mathcal{A}'n})$, so each sub-tour has length $\Theta(\sqrt{(\mathcal{A}/M)(K/M)}) = \Theta(\sqrt{\mathcal{A}K}/M)$, giving $t_{\text{loop}}(M) = \Theta(1/M)$. Convexity and monotonicity of $1/M$ carry over to the leading term, and we take them as the modeling assumption above.

Lemma 1 (Monotone, saturating charging wait): In the finite-source queue $M/M/c/M$, the mean wait $W_q(M, c)$ is non-decreasing in the population M , is strictly increasing once $M > c$, and satisfies the bound

$$0 \leq W_q(M, c) \leq \frac{M - c}{c\mu}, \quad M > c. \quad (14)$$

Proof: Let $\{n_t\}$ denote the birth-death chain of Section III on states $n \in \{0, \dots, M\}$ (UAVs at the station), with up-rates $(M - n)\lambda$ and down-rates $\min(n, c)\mu$ [35]. Increasing the population from M to $M + 1$ adds one source: at every state the up-rate weakly increases (from $(M - n)\lambda$ to $(M + 1 - n)\lambda$) while the down-rate is unchanged. A standard monotone coupling of the two chains keeps the $(M + 1)$ -chain pathwise at least as high as the M -chain, so the stationary number in system is stochastically increasing in M .

We convert this into a statement about the wait through the arriving-UAV view, rather than through the queue length alone: because the effective throughput $\lambda_{\text{eff}}(M)$ also grows with M , monotonicity of the ratio $W_q = L_q/\lambda_{\text{eff}}$ does not follow from monotonicity of L_q by itself. By the arrival theorem for finite-source (machine-repair) queues [35], a UAV that requests a charge in the M -population system finds the station distributed as the *stationary* distribution of the $(M - 1)$ -population system. A UAV that finds n present is served after $(n - c + 1)^+$ successive departures, each exponential of rate $c\mu$ while the c servers stay busy, so its conditional mean wait is $(n - c + 1)^+/(c\mu)$, which is non-decreasing in n . Averaging a non-decreasing function against a distribution

that is stochastically increasing in M , $W_q(M, c)$ is non-decreasing in M (and equals the Little's-law form L_q/λ_{eff} evaluated in Section III). The increase is strict once $M > c$, because states $n > c$ then carry positive probability in the arriving-UAV distribution and contribute a strictly positive wait. For (14), an arriving UAV finds at most $M - 1$ others present, so $n \leq M - 1$ and its conditional wait is at most $(M - 1 - c + 1)/(c\mu) = (M - c)/(c\mu)$; taking the expectation gives $W_q(M, c) \leq (M - c)/(c\mu)$. ■

Two features of Lemma 1 drive the result. Below c the wait is essentially zero (no UAV ever queues), and above c it climbs, ultimately at the linear rate $\Theta(M/(c\mu))$ of (14): its forward differences $\Delta W_q(M) \triangleq W_q(M+1, c) - W_q(M, c)$ are non-negative and rise from $\Delta W_q \approx 0$ in the underloaded region toward the saturation slope $1/(c\mu)$. We take this saturating shape as an explicit assumption.

Assumption 2 (Saturating charging increments): For each fixed c , the increment $\Delta W_q(M, c)$ is non-decreasing in M (so $W_q(\cdot, c)$ is discretely convex), and $\Delta W_q(M, c)$ is non-increasing in c .

Both parts reflect the same mechanism: added contention (larger M) steepens the wait, while added ports (larger c) flatten it and defer saturation. They are verified numerically across the tested field and capacity configurations in Section VI.

Proposition 1 (Congestion threshold and provisioning law): Under Assumption 1 and Lemma 1, the peak AoI $\text{AoI}(M)$ in (13) is quasi-convex in the swarm size M and admits a unique interior minimizer

$$M^* = \arg \min_{M \in \{1, 2, \dots\}} \text{AoI}(M), \quad (15)$$

so that adding UAVs strictly reduces AoI for $M < M^*$ (coverage dominates) and strictly increases it for $M > M^*$ (charging congestion dominates). Moreover M^* is non-decreasing in the port capacity c : enlarging a station never lowers the optimal swarm size.

Proof: Since c_0 is constant in (13), work with the forward difference

$$\Delta \text{AoI}(M) = \Delta c_1(M) + \Delta W_q(M), \quad (16)$$

where $\Delta c_1(M) = c_1(M+1) - c_1(M)$. By Assumption 1, c_1 is strictly decreasing and convex, so $\Delta c_1(M) < 0$ and $\Delta c_1(M)$ is non-decreasing (it rises toward 0 as $M \rightarrow \infty$). By Lemma 1 $\Delta W_q(M) \geq 0$, and by Assumption 2 it is non-decreasing. Hence the right-hand side of (16) is a sum of two non-decreasing terms, so $\Delta \text{AoI}(M)$ is non-decreasing in M : the sequence $\text{AoI}(M)$ is discretely convex.

It remains to check the endpoints of the sign of ΔAoI . For small M (in particular $M \leq c$) the queue is empty, $\Delta W_q(M) = 0$, while $\Delta c_1(M) < 0$, so $\Delta \text{AoI}(M) < 0$: AoI is still falling. As $M \rightarrow \infty$, $\Delta c_1(M) = \Theta(1/M) - \Theta(1/(M+1)) \rightarrow 0^-$ while, by (14), $\Delta W_q(M) \rightarrow 1/(c\mu) > 0$, so $\Delta \text{AoI}(M) > 0$ eventually: AoI is rising. A non-decreasing difference sequence that is negative for small M and positive for large M changes sign exactly once, from negative to positive. The last index with $\Delta \text{AoI}(M) < 0$ is the unique interior minimizer M^* , which establishes the U-shape.

For the provisioning law, fix M and increase c . The same monotone coupling, now applied to the down-rates $\min(n, c)\mu$ (which increase with c), makes the stationary occupancy stochastically smaller, so $W_q(M, c)$ is decreasing in c (more ports serve the same population); by Assumption 2 its increments $\Delta W_q(M; c)$ are non-increasing in c as well. Enlarging the station thus both lowers the wait and defers the onset of the steep saturation region to a larger population. The coverage difference $\Delta c_1(M)$ is unchanged. Hence for $c' > c$ we have $\Delta \text{AoI}(M; c') \leq \Delta \text{AoI}(M; c)$ for all M , so the single sign change of $\Delta \text{AoI}(\cdot; c')$ occurs at an index at least as large as that of $\Delta \text{AoI}(\cdot; c)$, i.e. $M^*(c') \geq M^*(c)$. ■

Proposition 1 says that congestion is a genuine, capacity-limited effect rather than an artifact of poor routing: even the balanced single-cell wait of Lemma 1 forces a finite M^* . The provisioning law $M^*(c)$ non-decreasing is the design counterpart: to operate a larger swarm at low AoI, one must add ports, not merely UAVs. Both the U-shape and the rightward shift of M^* with capacity are confirmed empirically, under optimal (load balancing) assignment, in Section VI.

B. Placement inversion under contention

For a single UAV the wait term in (13) vanishes ($M = 1 \leq c$), so $\text{AoI} = c_1(1) + 2 \text{dist}/V + \tau_{\text{charge}}$ depends on placement only through the recharge detour dist . The AoI-optimal station therefore sits where the detour is smallest, i.e. near the trajectory centroid: placement is coverage-driven. The next result shows this rule is not merely loose but can be strictly wrong for a swarm.

Proposition 2 (Placement inversion): There exist instances in which the coverage-optimal (minimum-detour) placement is strictly AoI-suboptimal for a swarm. Concretely, with a per-station port cap and two candidate sites, opening only the nearest site is beaten by opening both sites and splitting the swarm, for every swarm size above a finite threshold M_0 .

Proof: Consider two candidate sites and a per-station port cap $C_{\text{port}} = c$ (Section III), so a single station can host at most c ports; adding capacity beyond c requires opening a second station. Site 1 is central with zero extra detour; site 2 is offset, adding a round-trip detour $\delta = 2\text{dist}/V > 0$ for any UAV assigned to it. Two placements are compared under the same swarm and field, so the coverage term $c_1(M) = t_{\text{loop}}$ is identical in both:

$$(A) \text{ coverage-optimal: } \text{AoI}_A(M) = t_{\text{loop}} + W_q(M, c) + c_0, \quad (17)$$

$$(B) \text{ split: } \text{AoI}_B(M) = t_{\text{loop}} + \delta + W_q\left(\frac{M}{2}, c\right) + c_0, \quad (18)$$

where (A) sends all M UAVs to the near station (its c ports serving population M), and (B) opens both stations with c ports each and assigns $M/2$ UAVs to each; the peak in (B) is attained by the UAVs on the offset station, which carry the extra detour δ . Subtracting,

$$\text{AoI}_A(M) - \text{AoI}_B(M) = \left[W_q(M, c) - W_q\left(\frac{M}{2}, c\right) \right] - \delta. \quad (19)$$

Two bounds make the bracket diverge. For the near station, the throughput cannot exceed the service ceiling, $\lambda_{\text{eff}} \leq c\mu$; combined with the finite-source cycle identity $\lambda_{\text{eff}} = M/(T_{\text{up}} + W_q + \tau_{\text{charge}})$ (each of the M UAVs alternates operating, waiting, and charging), this gives the linear lower bound

$$W_q(M, c) \geq \frac{M}{c\mu} - T_{\text{up}} - \tau_{\text{charge}}. \quad (20)$$

For the split station, the Lemma upper bound (14) gives $W_q(M/2, c) \leq (M/2 - c)/(c\mu) \leq M/(2c\mu)$. Subtracting,

$$W_q(M, c) - W_q\left(\frac{M}{2}, c\right) \geq \frac{M}{2c\mu} - T_{\text{up}} - \tau_{\text{charge}}, \quad (21)$$

which grows without bound in M (the constants T_{up} and τ_{charge} do not depend on M), whereas δ is fixed. Therefore there is a finite threshold

$$M_0 = \min \left\{ M : W_q(M, c) - W_q\left(\frac{M}{2}, c\right) > \delta \right\} \quad (22)$$

such that $\text{AoI}_A(M) > \text{AoI}_B(M)$ for all $M > M_0$. For such M the minimum-detour placement (A) is strictly AoI-suboptimal, so the AoI-optimal placement must open the offset site and balance the charging load, even at the cost of a longer detour. ■

Proposition 2 formalises the inversion: as the swarm grows, the optimal placement stops chasing the shortest detour and starts spreading stations to balance queueing load, so the objective flips from coverage-driven to traffic-driven. The single-UAV placement rule of the base model is thus not a conservative approximation but the wrong optimizer for a contended swarm. This crossover, including the emergence of the load-balanced placement as the field and swarm scale, is demonstrated empirically in Section VI.

V. JOINT SOLVER

Problem (P0) in (12) couples binary placement, integer port capacity, continuous trajectories, a combinatorial assignment, and the nonconvex finite-source wait (10). Rather than attack this MINLP monolithically, we decompose it into blocks that are each tractable and solve them in a loop: a trajectory block, a finite-source queue evaluation, a port-allocation block, an assignment block, and an outer placement search. Fig. 2 sketches the flow. The blocks share the peak-AoI objective (11), so a change accepted by one block is scored by the same criterion the others use: the assignment search drives the peak AoI down monotonically to a local optimum, while the placement and port searches are enumerated exhaustively and retain the best configuration found. We describe each block, then the overall algorithm.

A. Trajectory block (GPU CETSP)

A UAV collects from any sensor within a horizontal radius r_c , so a patrol need not visit every sensor point: it need only pass within r_c of each assigned sensor. The shortest patrol is therefore the shortest closed tour of hover points whose r_c -disks cover the UAV's sub-field, a Close-Enough Traveling Salesman Problem (CETSP) [24]. We solve it by differentiable optimization. For UAV m , let $\mathbf{p}_{m,1}, \dots, \mathbf{p}_{m,n_m}$ be its hover

points, initialized at the k -means centroids of the sub-field. Within a cyclic order the closed-tour length is differentiable in the coordinates, so we minimize it by gradient descent. To also optimize the visiting order, and not freeze it at initialization, we alternate the continuous and combinatorial parts: we run a block of gradient steps on the coordinates, then reorder the current hover points by a 2-opt improvement of the tour, then repeat. This is a coordinate descent between the continuous positions and the discrete sequence, so the order adjusts as the positions move. The objective is the total closed-tour length plus a coverage penalty,

$$\mathcal{L} = \sum_{m \in \mathcal{M}} \left[\underbrace{\sum_{j=1}^{n_m} \|\mathbf{p}_{m,j+1} - \mathbf{p}_{m,j}\|}_{\text{tour length}} + \underbrace{\lambda_{\text{cov}} \sum_{k \in \mathcal{K}_m} [d_{m,k}^{\min} - r_c]_+^2}_{\text{coverage violation}} \right], \quad (23)$$

where indices are cyclic ($\mathbf{p}_{m,n_m+1} \equiv \mathbf{p}_{m,1}$), $\mathcal{K}_m$ is the sensor set assigned to UAV m , $d_{m,k}^{\min}$ is the distance from sensor k to its nearest hover point, $[\cdot]_+ = \max(0, \cdot)$, and λ_{cov} weights coverage against length. We minimize (23) by Adam over the hover-point coordinates, interleaved with the 2-opt reordering above. All UAVs of a scenario are stacked into one padded, masked batch and the gradient steps run in parallel on a GPU, so the whole swarm (and, by stacking seeds, whole sweeps) is optimized together. The block returns each UAV's optimized tour length, hence its flight time $t_{\text{fly}}^{\text{loop}}$ and, through (4), its revisit period t_{loop} and flight budget τ_{fly} . Because the sub-fields are fixed by the field partition, this block is independent of placement and is solved once at the start.

B. Charging queue and port allocation

Given an assignment, each open station s is the finite-source queue of Section III. Its mean wait $W_q(N_s, c_s)$ follows the machine-repair balance equations (8) to (10) from the number of assigned UAVs N_s , the ports c_s , the operating time (6), and the charge time τ_{charge} ; this evaluation is what turns a candidate configuration into a peak AoI through (11). Ports are then allocated across the open stations to reduce the peak wait under the budget $\sum_s c_s \leq C_{\text{tot}}$. Since W_q is convex-decreasing in the port count, the peak-reducing allocation adds ports where they buy the most, that is, to the most contended stations: a water-filling allocation. We realize it by scanning the integer port splits that sum to C_{tot} and keeping the split that yields the lowest peak AoI, which is exact for the small station budgets S_{max} considered here.

C. Assignment

With placement and ports fixed, UAVs are assigned to open stations to balance the queueing load. Starting from the nearest-station assignment, we run a greedy load-balancing local search: scanning the UAVs in turn, we apply the first single-UAV reassignment that strictly reduces the peak AoI (11), restart the scan after each accepted move, and stop

Algorithm 1 Block-coordinate joint solver for (P0)

Require: sensors $\mathcal{K}$, candidate sites $\mathcal{S}$, station budget $S_{\max}$, port budget C_{tot} , swarm size M

Ensure: placement $\mathbf{x}$, ports $\mathbf{c}$, assignment $\mathbf{a}$, trajectories $\{\mathbf{q}_m\}$, peak AoI $\bar{A}$

- 1: partition $\mathcal{K}$ into M sub-fields
- 2: **trajectory block:** minimize (23) by Adam on the GPU for all UAVs; obtain $t_{\text{loop},m}$, $\tau_{\text{fly},m}$
- 3: $\bar{A}^* \leftarrow +\infty$
- 4: **for** each site subset $\mathbf{x}$ of size $S_{\max}$ in $\mathcal{S}$ **do**
- 5: **for** each port split $\mathbf{c}$ with $\sum_s c_s = C_{\text{tot}}$ **do**
- 6: $\mathbf{a} \leftarrow$ nearest-station assignment
- 7: **repeat**
- 8: scan single-UAV reassignments; apply the first one that strictly reduces the peak AoI (11) (evaluating W_q by (10)) and restart the scan
- 9: **until** no single reassignment reduces the peak AoI
- 10: $\bar{A} \leftarrow$ peak AoI of $(\mathbf{x}, \mathbf{c}, \mathbf{a})$
- 11: **if** $\bar{A} < \bar{A}^*$ **then**
- 12: $\bar{A}^* \leftarrow \bar{A}$; store $(\mathbf{x}, \mathbf{c}, \mathbf{a})$
- 13: **end if**
- 14: **end for**
- 15: **end for**
- 16: **return** stored $(\mathbf{x}, \mathbf{c}, \mathbf{a})$, $\{\mathbf{q}_m\}$, $\bar{A}^*$

when no single reassignment improves it. Because the peak AoI is set by the worst station, driving it down repeatedly pulls UAVs off the most contended station onto lighter ones, so the search approximates the min-max (load-balancing) assignment $\arg \min_a \max_s W_q(N_s, c_s)$ without enumerating all assignments.

D. Placement (outer search)

The open-site subset $\mathbf{x}$ is chosen by an outer search over the candidate combinations. Candidate sites are the interior nodes of a regular grid over the field, and the search enumerates the $\binom{S}{S_{\max}}$ subsets of size $S_{\max}$. For each subset, the inner blocks (port allocation, assignment, and the queue evaluation) are run and the resulting peak AoI recorded; the subset with the lowest peak AoI is kept. This outer search is where the placement flips from coverage-driven to traffic-driven: a subset that spreads stations to split the charging load can beat the minimum-detour subset once contention is priced in through W_q .

E. Overall algorithm

Algorithm 1 assembles the blocks. The trajectory block is solved once, after which the placement, port, and assignment blocks are alternated: for each candidate placement the port split and the load-balancing assignment are optimized against the same peak-AoI objective, each evaluated through the finite-source queue.

Complexity and convergence. The trajectory block costs one batched GPU optimization independent of the placement search. The outer loops evaluate $\binom{S}{S_{\max}}$ placements times the number of port splits, and each evaluation runs the greedy

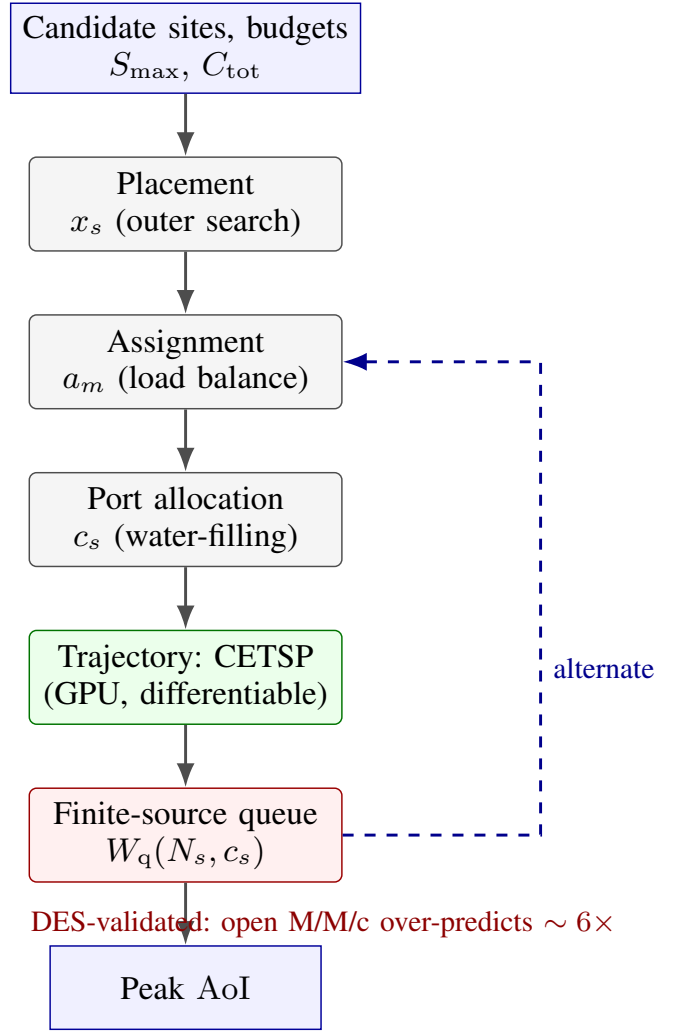


Fig. 2. Block-coordinate solver. The trajectory block (GPU CETSP) and the finite-source queue are alternated with port allocation, assignment, and an outer placement search.

assignment, whose local search terminates because every accepted move strictly lowers the peak AoI (a quantity bounded below by zero) and there are finitely many assignments. For a fixed placement and port split, the returned assignment is a local optimum of (11) under single-UAV moves; because the placement and port searches are exhaustive over the candidate grid and the integer splits, and the best configuration is retained, the solver returns the lowest peak AoI it evaluates. We do not claim global optimality: the trajectory order is fixed and the assignment search is greedy, so the result is the best configuration reachable by this search from the nearest-station initialization, not necessarily the global optimum of (P0).

VI. EXPERIMENTS**A. Setup**

We simulate $K=60$ sensors uniformly placed in a square field (side length varied from 5 to 20 km), a swarm of M rotary-wing UAVs cruising at $V=12$ m/s at altitude $H=100$ m, and a battery of about 67 Wh with a 15% reserve. The charge time per port is $\tau_{\text{charge}}=8$ min. The propulsion power follows

the rotary-wing model [36], and the air-to-ground link uses a line-of-sight free-space rate. Candidate stations lie on a 3×3 grid; unless stated otherwise the budget is $S_{\max}=2$ open stations and $C_{\text{tot}}=4$ ports. Every reported value is a mean over 20 to 30 random sensor fields (fixed seeds), with the standard deviation shown where relevant. All trajectory optimization runs on an NVIDIA RTX 4090.

B. Queue-model validation against a discrete-event simulation

The whole analysis rests on the charging-queue model, so we first validate it. We built a discrete-event simulation (DES) of the actual cycling UAVs, with a first-in-first-served station and c ports, in which each UAV alternates between patrolling for a randomized operating time and traveling to the station to recharge. The simulation supports both exponential and near-deterministic operating and service times, and we run each configuration over eight seeds and a long horizon after a warm-up, then measure the true mean queue wait. Fig. 3 compares three predictions against the DES. Under matched assumptions (exponential operating and service), the finite-source $M/M/c//N$ formula matches the DES to a mean relative error of 4.2%, which confirms the formula. For the realistic system (near-deterministic operating and charging), the DES wait is lower, so the finite-source model is a *conservative upper bound* (it exceeds the real wait in 23/23 configurations). In contrast, the open $M/M/c$ queue overpredicts the DES wait by about $6\times$ (a mean factor of 5.6), because it ignores the closed population: only M UAVs exist, so the arrival rate falls as more of them are already charging. Using the open queue therefore distorts the design decisions (we show below that it produces a spurious placement crossover), which is why we adopt the finite-source model throughout. The finite-source $M/M/c//N$ model assumes exponential service and is thus a conservative bound: it exceeds the real wait by about $4.5\times$. Replacing the exponential service by a deterministic-service $M/D/c//N$ approximation, obtained with the standard service-variability correction (the wait scales with $(1 + c_s^2)/2$, where c_s is the squared coefficient of variation of the service time, so $c_s^2=0$ halves the wait), tightens this to about $2.3\times$ (Fig. 3). The residual gap is due to the near-deterministic *operating* process, which a service-only correction does not capture; since both models are upper bounds, the reported AoI values remain safe rather than optimistic.

C. Proposition 1: congestion threshold and provisioning

Fig. 4 plots the peak AoI against swarm size M for several port budgets. Each curve is U-shaped: increasing M first lowers AoI (denser coverage) and then raises it (charging congestion), confirming the congestion threshold of Proposition 1. The optimal swarm size M^* grows with the charging capacity, as summarized in Table II, which is exactly the provisioning law: more ports support a larger cost-effective swarm.

D. Proposition 2: placement inversion

We compare, per instance, the coverage-optimal placement (minimum recharge detour, contention-blind) against the

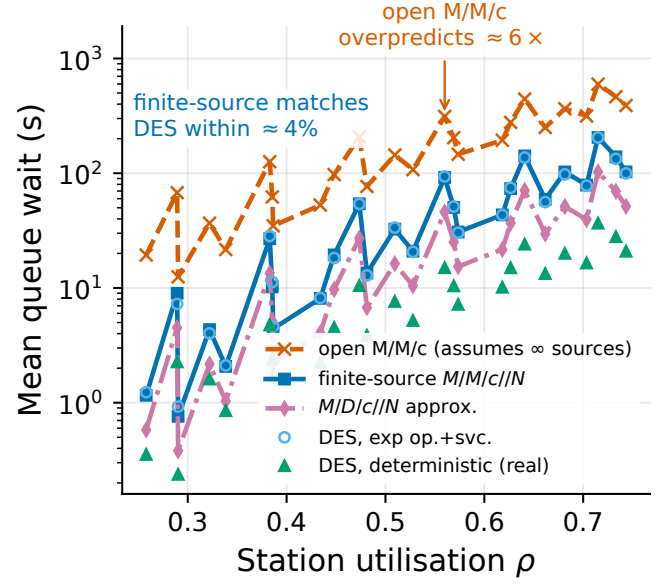


Fig. 3. Queue-model validation against discrete-event simulation. The finite-source model matches the DES under matched (exponential) assumptions and is a conservative upper bound for the real near-deterministic system, whereas the open $M/M/c$ overpredicts the wait about $6\times$.

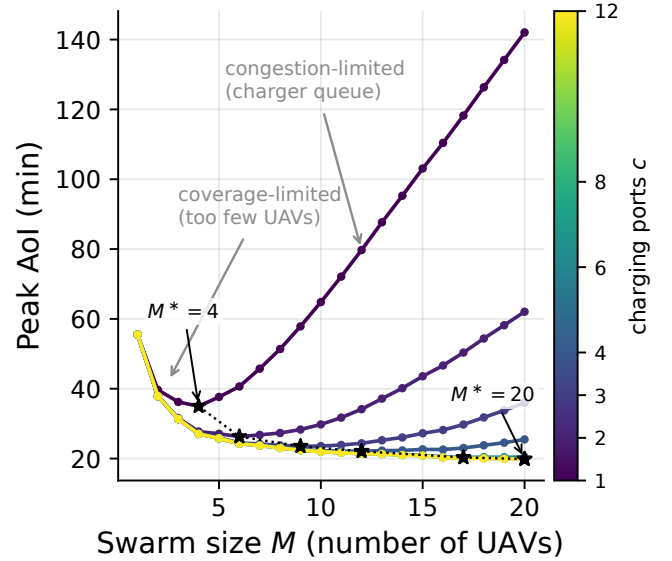


Fig. 4. Peak AoI is non-monotone in the swarm size M (finite-source model). Each port budget has an interior optimal swarm size M^* (black markers).

traffic-optimal placement (which balances the queueing load). The traffic-optimal placement lowers the peak AoI in 19/20 instances and never increases it, by 14.8% on average, and it selects *different* station sites in 18/20 instances (Fig. 6). This confirms Proposition 2: under contention the AoI-optimal placement inverts from coverage-driven to traffic-driven.

E. Joint solver versus baselines

We compare the proposed joint solver (CETSP trajectory with traffic-optimal placement) against: a single pooled sta-

TABLE II
PROVISIONING LAW: OPTIMAL SWARM SIZE M^* VERSUS PORT BUDGET c .

Ports c	1	2	3	4	6	8	12
M^*	4	6	9	12	17	20	20

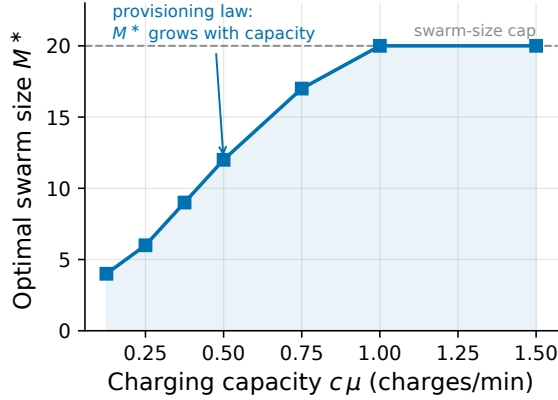


Fig. 5. Provisioning law: the optimal swarm size M^* increases with the charging capacity, so a larger port budget supports a larger cost-effective swarm.

tion (all ports at one central station, the strong Wei-style baseline [3]), a coverage-optimal placement, a round-robin assignment, and an ablation that replaces the CETSP trajectory with a nearest-neighbor patrol. Table III reports the ablation at a 15 km field: the proposed method attains the lowest peak AoI, the CETSP trajectory contributes about 3.9%, and the contention-aware placement contributes substantially more (about 18% over the coverage-optimal placement). A Wilcoxon signed-rank test gives $p=1.9 \times 10^{-6}$ against the closest competitor, the no-CETSP ablation, so the improvement is statistically significant.

Fig. 8 sweeps the field size. The proposed distributed, contention-aware placement beats the single pooled station at every field size, and the gain grows with the field (from 4.1% at 5 km to 12.2% at 20 km) because travel cost dominates as the field grows. The choice of queue model is decisive here. Had the charging system been modeled as an open $M/M/c$ queue, a spurious crossover would appear in which the single pooled station wins in small fields, because the open queue overpenalizes split stations. The DES-validated finite-source model, whose wait saturates rather than diverging, removes this artifact, and the distributed placement is preferred throughout.

F. Scaling to a large swarm

A swarm is only meaningful at larger scale, so we scale the fleet to $M = 60$ UAVs over a 20 km field with $K = 8M$ sensors, and we grow the charging capacity with the swarm ($C_{\text{tot}} = \lceil M/3 \rceil$ ports) as the provisioning law of Proposition 1 requires. Fig. 9 shows that the peak AoI stays controlled as the swarm grows (from 50.6 min at $M = 20$ to 47.6 min at $M = 60$), because the sensors and the charging capacity scale together, and that the proposed contention-aware placement

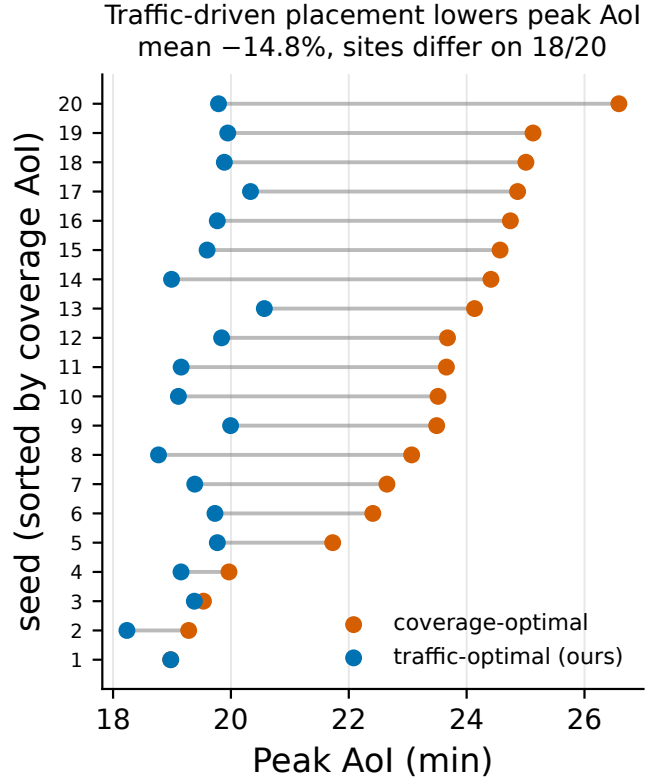


Fig. 6. Placement inversion. Traffic-optimal placement (which balances the queueing load) beats the coverage-optimal (minimum-detour) placement on every seed, and chooses different sites on 18/20 seeds.

TABLE III
ABLATION AT A 15 KM FIELD ($M=12$, MEAN PEAK AOI IN MINUTES).

Method	Peak AoI (min)
Proposed (CETSP + traffic-optimal)	41.1
No CETSP (nearest-neighbor patrol)	42.8
Single pooled station [3]	46.2
Coverage-optimal placement	50.4
Round-robin assignment	56.7

keeps its advantage over the single pooled station and the coverage-optimal placement across the whole range, by 10.2 to 12.7%. The design therefore applies to genuine swarms, not only to small fleets.

G. Solver optimality

The placement and assignment blocks are greedy, so we ask how far they are from the true optimum. On small instances the exact optimum is computable by enumerating every open-site subset, every port split, and every UAV-to-station assignment; we run the greedy solver and the enumeration on identical per-UAV states, so the gap measures only the combinatorial heuristic. Table IV reports the result over 15 seeds. The greedy placement and assignment are *exactly* optimal for $M \leq 6$ (they match the enumerated optimum on every seed), and remain within 0.11% on average and below 1% in the worst case at $M = 8$. The heuristic therefore matches the optimum wherever

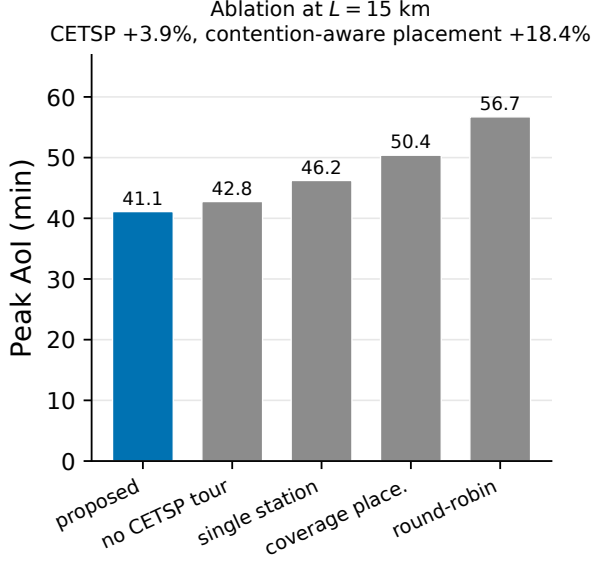


Fig. 7. Ablation at a 15 km field. Both the contention-aware placement and the CETSP trajectory contribute to the peak-AoI reduction of the proposed method.

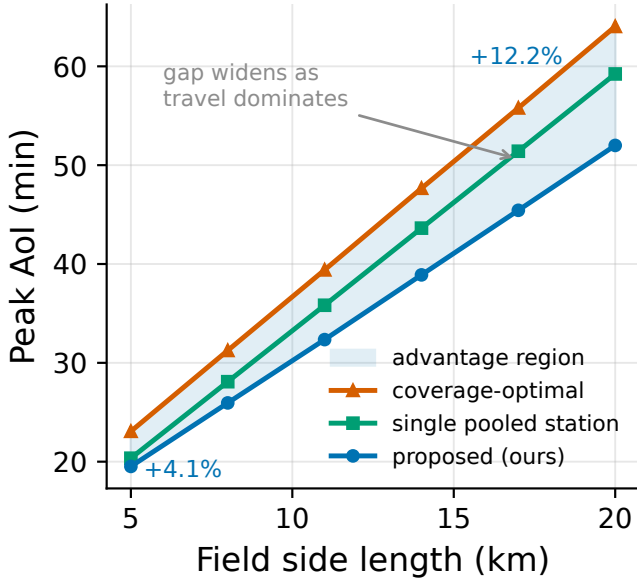


Fig. 8. Field-size sweep. The proposed distributed placement beats the single pooled station at all sizes, with a widening margin as travel cost grows.

an exact comparison is tractable, which supports its use at the larger swarm sizes where enumeration is infeasible.

H. Robustness

Fig. 10 sweeps the main scenario parameters at $L = 14$ km. The gain of the proposed placement over the single pooled station is stable across the collection radius (11.2 to 11.4% for r_c from 100 to 400 m) and the charge time (8.5 to 13.5%). It is also larger, not smaller, on a non-uniform field: when the sensors form clusters the gain rises to 19.0%, because concentrated recharge traffic makes load balancing more valuable.

Scaling to a large swarm (capacity grows with M)

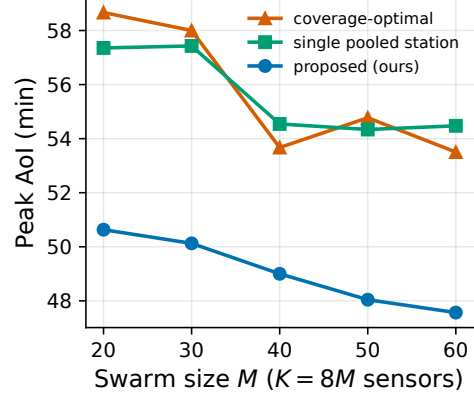


Fig. 9. Scaling to a large swarm (M up to 60, $K = 8M$ sensors, capacity $C_{\text{tot}} = \lceil M/3 \rceil$ growing with the swarm). The proposed placement keeps its advantage over the baselines at scale.

TABLE IV

OPTIMALITY GAP OF THE GREEDY SOLVER VERSUS THE EXACT OPTIMUM.

Swarm size M	3	4	5	6	7	8
Mean gap (%)	0	0	0	0	0.03	0.11
Optimal seeds	15/15	15/15	15/15	15/15	13/15	10/15

The advantage is therefore robust and, if anything, strengthens in realistic non-uniform deployments.

I. Discussion and design guidelines

The results translate into concrete guidance for deploying a UAV swarm for fresh data collection. First, the swarm should not be sized by coverage alone. Because the peak AoI is non-monotone in the swarm size, there is an optimal swarm size M^* beyond which extra UAVs only add charging congestion and worsen freshness; Table II shows that M^* is set by the installed charging capacity, so the swarm size and the number of ports must be provisioned together. When freshness targets tighten, adding ports (raising M^*) is often more effective than adding UAVs to an already congested system.

Second, charging stations should be placed to balance the queueing load, not only to minimize flight detour. The coverage-optimal placement concentrates the recharge traffic and inflates the wait; spreading the stations lowers the peak AoI even at the cost of longer trips, and the benefit grows with the field size because travel then dominates. A single pooled station remains a sensible choice only in small fields where travel is cheap, which the field-size sweep makes explicit.

Third, the queueing model matters as much as the optimizer. Treating the charging system as an open $M/M/c$ queue overstates the wait by about $6\times$ and, as shown above, can flip a design decision (the spurious small-field crossover). The finite-source model removes this bias and, being a conservative upper bound on the near-deterministic reality, yields safe rather than optimistic AoI estimates. We therefore recommend the finite-source model as the default planning tool for swarm recharging with a bounded number of vehicles.

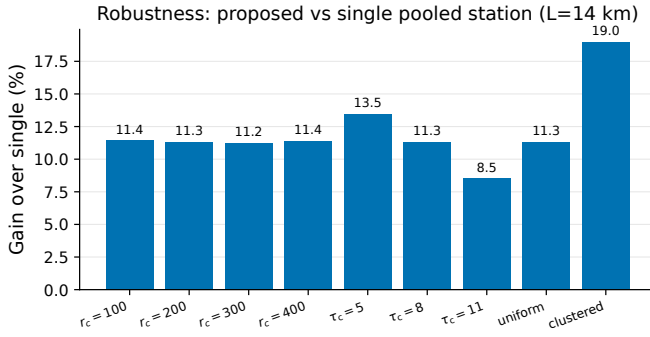


Fig. 10. Robustness of the proposed method's gain over the single pooled station to the collection radius r_c , the charge time τ_{charge} , and a uniform versus clustered sensor field ($L = 14$ km, $M = 12$).

Finally, the ablation isolates where the gain comes from. The contention-aware placement provides the larger share, while the GPU close-enough-TSP trajectory adds a smaller but consistent improvement by shortening the revisit period at a realistic collection radius. Both components are complementary, and the joint solver combines them without violating the energy and budget constraints.

VII. CONCLUSION

We studied fresh data collection by a UAV swarm that recharges at a small number of shared, multi-port charging stations. We formulated the joint placement, capacity, trajectory, and assignment problem under a charging queue, and we showed that the correct queue model is a finite-source (machine-repair) queue, not an open $M/M/c$: a discrete-event simulation confirms the finite-source formula within about 4% and shows that the open queue overpredicts the wait by roughly $6\times$ and distorts the design. Two structural results followed: a congestion threshold that makes the AoI non-monotone in the swarm size, with a provisioning law linking the optimal swarm size to the charging capacity, and a placement inversion in which the AoI-optimal placement becomes traffic-driven rather than coverage-driven. A block-coordinate solver with a GPU close-enough-TSP trajectory optimizer realized these gains, reducing peak AoI across field sizes against strong baselines.

Several limitations remain. The finite-source model assumes exponential operating and service times; the deterministic-service $M/D/c/N$ approximation we introduce tightens the bound, but a residual gap remains from the near-deterministic operating process, which a full non-exponential treatment would close. The trajectory block uses a close-enough-TSP heuristic with 2-opt reordering rather than an exact solver; it matches the exact optimum on the small instances where enumeration is tractable, but tighter guarantees at scale are open. Finally, extending the model to stochastic data arrivals and online operation would move the design from planning toward real-time control.

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